

Differential Equations

Basic concepts and classification

Differential equation: An equation that contains an unknown function and one or more of its derivatives. Here y is the unknown function and x is the independent variable.

$$y' = \frac{dy}{dx}, \quad y'' = \frac{d^2y}{dx^2}, \quad \dots, \quad y^{(n)} = \frac{d^ny}{dx^n}.$$

Example: For $f(x, y) = x^2 + 3y^4 + 5$, the partial derivatives are:

$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 12y^3.$$

Ordinary and partial differential equations

- **Ordinary differential equation (ODE):** The unknown function depends on one independent variable.

$$y'' - xy' + 2y = 0, \quad \frac{dy}{dx} = 2(4 - x)(1 - x).$$

- **Partial differential equation (PDE):** The unknown function depends on two or more independent variables.

$$\frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial z}{\partial y} = 0, \quad \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}.$$

Order and degree

Order is the order of the highest derivative in the equation, not its exponent.

$$y' + y = x^2 \quad (\text{order } 1)$$

$$y'' + 3(y')^5 = 0 \quad (\text{order } 2)$$

$$\frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial z}{\partial y} = 0 \quad (\text{order } 2)$$

Degree: If a differential equation is polynomial in y and its derivatives, the degree is the power of the highest-order derivative.

$$(y''')^2 + (y'')^4 + y = x \quad (\text{order } 3; \text{degree } 2)$$

$$\left(\frac{dy}{dx}\right)^3 + 2x \frac{dy}{dx} = 2y + 1 \quad (\text{order } 1; \text{degree } 3)$$

$$xy''' + 2(y')^4 + y = 0 \quad (\text{order } 3; \text{degree } 1)$$

NOTE: If a differential equation is not polynomial in its derivatives, its degree is not defined.

$$\sin(y'') + y = 0 \quad (\text{order } 2; \text{degree undefined})$$

NOTE: Remove the radical before determining the degree:

$$y' = \sqrt{\frac{y}{x}} \Rightarrow (y')^2 = \frac{y}{x} \quad (\text{order } 1; \text{degree } 2)$$

Linear and nonlinear equations

An equation that is not linear is nonlinear.

An n th-order ODE is linear if it can be written as

$$a_{n(x)}y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = f(x), \quad a_{n(x)} \neq 0.$$

The $a_{i(x)}$ are coefficients that depend on x . A linear equation has the following properties:

1. y and all its derivatives must appear to the first power.
2. Products such as yy' and $y'y''$ must not occur.

3. Functions of y or its derivatives such as $\sin y$, e^y , and $\ln y$ must not occur.
4. The coefficients must depend only on x or be constants.

NOTE: Numerical coefficients are **constant coefficients**; functions of x are **variable coefficients**. If $f(x) = 0$, the equation is **homogeneous**; if $f(x) \neq 0$, it is **nonhomogeneous**.

Classification examples

Equation	Type	Order	Degree	Linearity	Coefficients	Homogeneity
$xy''' + 2(y')^4 + y = 0$	ODE	3	1	Nonlinear	Variable	Homogeneous
$y'' + 9y = \sin y$	ODE	2	1	Nonlinear	Constant	Homogeneous
$y' = \sqrt{\frac{y}{x}}$	ODE	1	2 definitely nonlinear	Nonlinear	Variable	Homogeneous
$(1 - x^2)y'' - xy' + 4y = 0$	ODE	2	1	Linear	Variable	Homogeneous
$y'' - xy' + 2y = 0$	ODE	2	1	Linear	Variable	Homogeneous
$\left(\frac{dy}{dx}\right)^3 + 2x\frac{dy}{dx} = 2y + 1$	ODE	1	3	Nonlinear	Variable	Nonhomogeneous
$\frac{dy}{dx} = 2(4 - x)(1 - x)$	ODE	1	1	Linear	Constant	Nonhomogeneous
$\frac{\partial^2 z}{\partial x^2} + 2\frac{\partial z}{\partial y} = 0$	PDE	2	1	Linear	Constant	Homogeneous

Solutions

A function $y = \varphi(x)$ is a **solution** of a differential equation on an interval I if it satisfies the equation for every $x \in I$.

Example: Show that $y = Ce^{2x}$ is a solution of $y' - 2y = 0$ for every C . Since $y' = 2Ce^{2x}$:

$$y' - 2y = 2Ce^{2x} - 2Ce^{2x} = 0.$$

Explicit and implicit solutions

A function $y = \varphi(x)$ is an **explicit solution** if substituting it for y satisfies the equation $F\left(x, y, \frac{dy}{dx}, \dots, \frac{d^ny}{dx^n}\right) = 0$. An implicit solution has the form $g(x, y) = 0$.

General and particular solutions; initial value problems

A solution with n arbitrary constants is a **general solution**. Assigning values to the constants gives **particular solutions**.

Initial value problem (IVP): A differential equation together with an initial condition such as $y(x_0) = y_0$.

Example: For $y' = 2y$ and $y(0) = 3$, the general solution is $y = Ce^{2x}$. The initial condition gives $C = 3$, so the particular solution is:

$$y = 3e^{2x}.$$

Singular solution

A **singular solution** cannot be obtained from the general solution by any choice of its arbitrary constants.

Example: The general solution of $y = xy' + (y')^2$ is given as $y = Cx + C^2$. The function $y = -\frac{x^2}{4}$ is a singular solution: $y' = -\frac{x}{2}$, and:

$$x\left(-\frac{x}{2}\right) + \left(-\frac{x}{2}\right)^2 = -\frac{x^2}{4} = y.$$

Deriving differential equations from families of curves

Let c be an arbitrary constant and consider the family

$$f(x, y, c) = 0 \quad (1)$$

Differentiate with respect to x , treating $y = y(x)$:

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} = 0 \quad (2)$$

Eliminating c from (1) and (2) gives the differential equation

$$F(x, y, y') = 0.$$

NOTE: With two constants, $f(x, y, c_1, c_2) = 0$, take two derivatives. The resulting three equations eliminate the two constants to give

$$F(x, y, y', y'') = 0.$$

Example 1.1.1 - $y = ax^2 + b$

Let a and b be arbitrary constants. Find the lowest order differential equation whose general solution is $y = ax^2 + b$.

(two constants $\Rightarrow$ two derivatives) Differentiate twice:

$$y' = 2ax, \quad y'' = 2a.$$

Substitute $a = \frac{y''}{2}$ to obtain the differential equation:

$$y' = xy''$$

Example 1.1.2

$$y = a_1 + a_2x + a_3x^2 + a_4x^3.$$

(g.s.). Find the lowest order differential equation.

$$y' = a_2 + 2a_3x + 3a_4x^2, \quad y'' = 2a_3 + 6a_4x, \quad y''' = 6a_4.$$

$$y^{(4)} = 0$$

Example 1.1.3 - $y = Cx + C^2 + C$ (g.s.)

Since $y' = C$, substitute y' for C :

$$y = xy' + (y')^2 + y'.$$

Example 1.1.4 - $y = C_1x + \frac{C_2}{x}$ (g.s.)

$$y' = C_1 - \frac{C_2}{x^2}, \quad y'' = 2\frac{C_2}{x^3}.$$

Thus $C_2 = x^3 \frac{y''}{2}$. Then $y' = C_1 - x \frac{y''}{2}$, so $C_1 = y' + x \frac{y''}{2}$. Substituting into the original equation gives:

$$y = x \left(y' + x \frac{y''}{2} \right) + x^2 \frac{y''}{2} = xy' + x^2 y''.$$

$$x^2 y'' + xy' - y = 0$$

Example 1.1.7 - $y = e^{Cx + \frac{x}{C}}$

For $\ln y = Cx + \frac{x}{C}$:

$$\frac{y'}{y} = C + \frac{1}{C}, \quad \frac{\ln y}{x} = C + \frac{1}{C}.$$

Equating these expressions gives:

$$\boxed{xy' - y \ln y = 0}$$

★ **Example 1.1.8** - $y = xe^{\frac{C}{x}}$

$$y' = e^{\frac{C}{x}} + xe^{\frac{C}{x}} \left(-\frac{C}{x^2} \right) = e^{\frac{C}{x}} \left(1 - \frac{C}{x} \right).$$

$$\frac{y}{x} = e^{\frac{C}{x}} \Rightarrow \ln \left(\frac{y}{x} \right) = \frac{C}{x}.$$

$$\boxed{y' = \frac{y}{x} \left(1 - \ln \left(\frac{y}{x} \right) \right)}$$

★ **Example 1.1.9** - $y = C_1 e^{-x} + C_2 e^{7x}$

$$y = C_1 e^{-x} + C_2 e^{7x} \quad (1)$$

$$y' = -C_1 e^{-x} + 7C_2 e^{7x} \quad (2)$$

$$y'' = C_1 e^{-x} + 49C_2 e^{7x} \quad (3)$$

$$(1) + (2) \Rightarrow y' + y = 8C_2 e^{7x}.$$

$$(2) + (3) \Rightarrow y'' + y' = 56C_2 e^{7x}.$$

$$\frac{y' + y}{y'' + y'} = \frac{1}{7} \Rightarrow 7y' + 7y = y'' + y'.$$

$$\boxed{y'' - 6y' - 7y = 0}$$

Check: no arbitrary constants remain ✓; a second-order derivative appears ✓.

MATH201 · Differential Equations

Week 2 lecture notes

Families of curves (continued)

Example 1.1.10 — $y = C_1x + C_2e^{-x}$

$y' = C_1 - C_2e^{-x}$ and $y'' = C_2e^{-x}$, so $C_1 = y' + y''$ and

$$y = x(y' + y'') + y''.$$

Example 1.1.11 — $y = x^2 + C_1e^x + C_2e^{-x}$

Since $y'' = 2 + C_1e^x + C_2e^{-x}$, subtraction eliminates both constants:

$$y = x^2 + y'' - 2, \quad \text{equivalently} \quad y'' - y = 2 - x^2.$$

First-order differential equations

Separable differential equations

A first-order equation can be written as $d_y y = f(x, y)$, or in differential form as $M(x, y)dx + N(x, y)dy = 0$ (with $f = -\frac{M}{N}$ where $N \neq 0$). It is **separable** if it can be rearranged into $P(x)dx + Q(y)dy = 0$: collect the y terms with dy and the x terms with dx . Direct integration gives

$$\int P(x)dx + \int Q(y)dy = C.$$

When dividing by a factor involving x or y , check the excluded values against the equation.

Example 1.2.1

Solve $x(y^2 - 1)dx - y(x^2 - 1)dy = 0$ with $y(2) = 2$. On the region $x > 1, y > 1$ containing the initial point, separate:

$$\frac{xdx}{x^2 - 1} - \frac{ydy}{y^2 - 1} = 0, \quad \int \frac{xdx}{x^2 - 1} = \int \frac{ydy}{y^2 - 1}.$$

For the left integral, set $u = x^2 - 1$, so $du = 2xdx$:

$$\int \frac{xdx}{x^2 - 1} = \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln(x^2 - 1).$$

Similarly, set $v = y^2 - 1$, $dv = 2ydy$ on the right. Thus

$$\ln(x^2 - 1) - \ln(y^2 - 1) = C, \quad \frac{x^2 - 1}{y^2 - 1} = e^C = K.$$

Since e^C is also a constant, it may be renamed C ; here we use K to distinguish it, with $K > 0$ on this region. The initial condition gives $K = 1$, so $x^2 = y^2$. Since $x, y > 1$,

$$y = x.$$

The differential equation also has the constant solutions $y(x) = 1$ and $y(x) = -1$, excluded by division by $y^2 - 1$. Neither satisfies $y(2) = 2$.

Example 1.2.2

Solve $(1 + x^2 + y^2 + x^2y^2)dy = y^2dx$ with $y(0) = -1$. Group and factor:

$$1 + x^2 + y^2 + x^2y^2 = x^2(1 + y^2) + (1 + y^2) = (1 + x^2)(1 + y^2).$$

For $y \neq 0$, separate:

$$\frac{(1 + y^2)dy}{y^2} = \frac{dx}{1 + x^2}.$$

Integration yields $y - \frac{1}{y} = \arctan x + C$. From $y(0) = -1$, $C = 0$; therefore

$$y - \frac{1}{y} = \arctan x, \quad y(0) = -1.$$

The constant solution $y(x) = 0$ satisfies the equation but is lost when dividing by y^2 . It does not satisfy $y(0) = -1$.

Example 1.2.3

Solve $(x^2 - 2x + 1)dy - (y^2 + 2y + 1)dx = 0$ with $y(2) = 1$. The factors are $(x - 1)^2$ and $(y + 1)^2$. For $x \neq 1$ and $y \neq -1$, division and integration give

$$\int \frac{dy}{(y + 1)^2} - \int \frac{dx}{(x - 1)^2} = 0.$$

Thus $-\frac{1}{y+1} + \frac{1}{x-1} = C$. The initial condition gives $C = \frac{1}{2}$, so

$$-\frac{1}{y+1} + \frac{1}{x-1} = \frac{1}{2}.$$

The constant solution $y(x) = -1$ also satisfies the differential equation and is excluded by this division. It does not satisfy $y(2) = 1$.

Example 1.2.5

Solve $\tan^2 y dy = \sin^3 x dx$ with $y(0) = 0$. The equation requires $\cos y \neq 0$. Using $\tan^2 y = \sec^2 y - 1$,

$$\int \tan^2 y dy = \tan y - y.$$

For the right integral, write $\sin^3 x = \sin x(1 - \cos^2 x)$ and set $u = \cos x$, $du = -\sin x dx$:

$$\int \sin^3 x dx = -\int (1 - u^2) du = -u + \frac{u^3}{3} = -\cos x + \frac{\cos^3 x}{3}.$$

Hence $\tan y - y = -\cos x + \frac{\cos^3 x}{3} + C$. At $(0, 0)$, $C = \frac{2}{3}$, so

$$\tan y - y = -\cos x + \frac{\cos^3 x}{3} + \frac{2}{3}.$$

Example 1.2.6

Solve $xy^2(1+x)dy + (1+y^3)dx = 0$ with $y(1) = 1$. Division by $x(1+x)(1+y^3)$ requires $x \neq 0$, $x \neq -1$, and $y \neq -1$:

$$\frac{y^2 dy}{1+y^3} + \frac{dx}{x(1+x)} = 0.$$

Set $u = 1 + y^3$, so $du = 3y^2 dy$; for the other integral use partial fractions:

$$\int \frac{y^2 dy}{1+y^3} = \frac{1}{3} \int \frac{du}{u} = \frac{1}{3} \ln|1+y^3|, \quad \frac{1}{x(1+x)} = \frac{1}{x} - \frac{1}{1+x}.$$
$$\int \frac{dx}{x(1+x)} = \ln|x| - \ln|1+x| = \ln\left|\frac{x}{x+1}\right|.$$

Hence $\frac{1}{3} \ln|1+y^3| + \ln\left|\frac{x}{x+1}\right| = \ln K$, or $\frac{x(1+y^3)^{\frac{1}{3}}}{x+1} = K$ on the branch of the initial point. The condition gives $K = 2^{-\frac{2}{3}}$. Cubing yields

$$\frac{x^3(1+y^3)}{(x+1)^3} = \frac{1}{4}.$$

The differential equation also admits $y(x) = -1$, lost when dividing by $1+y^3$. It does not satisfy $y(1) = 1$.

Example 1.2.9

Solve $y' = x \sin \frac{x}{\tan y} y$ with $y(0) = \pi$. The equation requires both $\sin y \neq 0$ and $\cos y \neq 0$, so that $\tan y$ is defined and nonzero. Separation gives $\tan y dy = x \sin x dx$. For the left integral, set $t = \cos y$, $dt = -\sin y dy$:

$$\int \tan y dy = \int \frac{\sin y}{\cos y} dy = - \int \frac{dt}{t} = -\ln|\cos y|.$$

For the right integral, integration by parts with $u = x$, $dv = \sin x dx$, $du = dx$, and $v = -\cos x$ gives

$$\int x \sin x dx = -x \cos x + \int \cos x dx = -x \cos x + \sin x.$$

Therefore

$$-\ln|\cos y| = -x \cos x + \sin x + C.$$

At $y = \pi$, $\tan y = 0$, so the differential equation is undefined at the initial point. For a solution with $\lim_{x \rightarrow 0^+} y(x) = \pi$, taking the limit in the integrated relation gives $C = 0$. Define $A(x) = \sin x - x \cos x$; then

$$|\cos y| = e^{-A(x)}.$$

Near $y = \pi$, $\cos y < 0$, so $\cos y = -e^{-A(x)}$. Since $A(0) = 0$ and $A'(x) = x \sin x > 0$ for $0 < x < \pi$, we have $A(x) > 0$ there. There are two branches approaching π from opposite sides:

$$y_-(x) = \pi - \arccos(e^{-A(x)}), \quad y_+(x) = \pi + \arccos(e^{-A(x)}), \quad 0 < x < \pi.$$

Both satisfy the equation on this open interval and tend to π as $x \rightarrow 0^+$. For $-\pi < x < 0$, $A(x) < 0$, so $e^{-A(x)} > 1$: no real solution of the $C = 0$ relation exists there. Hence there is no real solution on $-\pi < x < 0$ that tends to π as $x \rightarrow 0^-$.

Both branches extend continuously to $y(0) = \pi$. The initial value problem has no classical solution at $x = 0$, because the right-hand side of the differential equation is undefined there.